In [1]:
```
!pip install requests pandas networkx matplotlib
```

```
Requirement already satisfied: requests in /opt/anaconda3/lib/python3.13/site-p
ackages (2.32.5)
Requirement already satisfied: pandas in /opt/anaconda3/lib/python3.13/site-pac
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Requirement already satisfied: matplotlib in /opt/anaconda3/lib/python3.13/sit
e-packages (3.10.6)
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ython3.13/site-packages (from requests) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /opt/anaconda3/lib/python3.13/si
te-packages (from requests) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /opt/anaconda3/lib/python
3.13/site-packages (from requests) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /opt/anaconda3/lib/python
3.13/site-packages (from requests) (2026.1.4)
Requirement already satisfied: numpy>=1.26.0 in /opt/anaconda3/lib/python3.13/s
ite-packages (from pandas) (2.3.5)
Requirement already satisfied: python-dateutil>=2.8.2 in /opt/anaconda3/lib/pyt
hon3.13/site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /opt/anaconda3/lib/python3.13/si
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Requirement already satisfied: tzdata>=2022.7 in /opt/anaconda3/lib/python3.13/
site-packages (from pandas) (2025.2)
Requirement already satisfied: contourpy>=1.0.1 in /opt/anaconda3/lib/python3.1
3/site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /opt/anaconda3/lib/python3.13/si
te-packages (from matplotlib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in /opt/anaconda3/lib/python
3.13/site-packages (from matplotlib) (4.60.1)
Requirement already satisfied: kiwisolver>=1.3.1 in /opt/anaconda3/lib/python
3.13/site-packages (from matplotlib) (1.4.9)
Requirement already satisfied: packaging>=20.0 in /opt/anaconda3/lib/python3.1
3/site-packages (from matplotlib) (25.0)
Requirement already satisfied: pillow>=8 in /opt/anaconda3/lib/python3.13/site-
packages (from matplotlib) (12.0.0)
Requirement already satisfied: pyparsing>=2.3.1 in /opt/anaconda3/lib/python3.1
3/site-packages (from matplotlib) (3.2.5)
Requirement already satisfied: six>=1.5 in /opt/anaconda3/lib/python3.13/site-p
ackages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

In [2]:
```
import requests
import pandas as pd
import networkx as nx
import matplotlib.pyplot as plt
import re
from itertools import combinations
from collections import Counter
from networkx.algorithms.community import greedy_modularity_communities
```

In [3]:
```
# Collect the newest 100 posts from r/MachineLearning
url = "https://www.reddit.com/r/MachineLearning/new.json?limit=100"
```

```python
headers = {
    "User-Agent": "network-analysis-project"
}

response = requests.get(url, headers=headers)

# Check whether request was successful
print("Status code:", response.status_code)

data = response.json()

posts = []

for post in data["data"]["children"]:
    posts.append({
        "post_id": post["data"]["id"],
        "title": post["data"]["title"],
        "author": post["data"]["author"],
        "created_utc": post["data"]["created_utc"],
        "num_comments": post["data"]["num_comments"],
        "score": post["data"]["score"]
    })

posts_df = pd.DataFrame(posts)

print("Total posts collected:", len(posts_df))
posts_df.head()
```

```
Status code: 200
Total posts collected: 100
```

Out[3]:

	post_id	title	author	created_utc	num_comments	score
0	1rtsbkv	[P] Karpathy's autoresearch with evolutionary ...	hgarud	1.773516e+09	0	10
1	1rtouwo	[D] Is anyone interested in the RL ↔ Neuroscie...	Kooky_Ad2771	1.773508e+09	11	1
2	1rtled3	[D] Need advice on handling a difficult ACL AR...	Aloo_Ka_Pakoda	1.773500e+09	2	5
3	1rtkazr	[D] ACL ARR 2026 Jan cycle — Does the commitme...	Distinct_Relation129	1.773497e+09	1	2
4	1rtjirw	The arXiv is separating from Cornell Universit...	Benlus	1.773495e+09	68	301

In [4]:

```
posts_df.to_csv("reddit_posts.csv", index=False)
print("Saved: reddit_posts.csv")
```

```
Saved: reddit_posts.csv
```

In [5]:

```
# Common words that do not represent meaningful discussion topics
stopwords = {
    "the", "is", "a", "an", "and", "or", "for", "to", "of", "in", "on", "with"
    "how", "what", "why", "best", "using", "use", "used", "from", "about",
    "my", "your", "this", "that", "are", "can", "into", "need", "help",
    "any", "all", "new", "after", "before", "more", "than", "when", "where",
    "which", "who", "should", "would", "could", "their", "there", "them",
    "some", "many", "much", "just", "over", "under", "also", "have", "has",
    "had", "was", "were", "being", "been", "do", "does", "did", "i", "we",
    "you", "they", "it", "our", "me", "us"
}

def extract_keywords(title):
    """
    Convert a Reddit title into a cleaned list of keywords.
    Steps:
    1. Lowercase the text
    2. Keep only alphabetic words
    3. Remove stopwords
```

```
    4. Remove very short words
    5. Remove duplicates within the same title
    """
    words = re.findall(r"\b[a-zA-Z]+\b", str(title).lower())
    words = [w for w in words if w not in stopwords and len(w) > 2]
    return sorted(list(set(words)))
```

In [6]:

```
# Extract keywords for each post
posts_df["keywords"] = posts_df["title"].apply(extract_keywords)

# Remove posts with no keywords
posts_df = posts_df[posts_df["keywords"].map(len) > 0].reset_index(drop=True)

print("Posts remaining after keyword extraction:", len(posts_df))
posts_df[["post_id", "title", "keywords"]].head(10)
```

```
Posts remaining after keyword extraction: 100
```

Out[6]:

	post_id	title	keywords
0	1rtsbkv	[P] Karpathy's autoresearch with evolutionary ...	[autoresearch, database, evolutionary, karpathy]
1	1rtouwo	[D] Is anyone interested in the RL ↔ Neuroscie...	[anyone, deep, dive, interested, neuroscience,...
2	1rtled3	[D] Need advice on handling a difficult ACL AR...	[acl, advice, arr, difficult, handling, situat...
3	1rtkazr	[D] ACL ARR 2026 Jan cycle — Does the commitme...	[acl, arr, chosen, commitment, cycle, during, ...
4	1rtjirw	The arXiv is separating from Cornell Universit...	[arxiv, ceo, cornell, decades, establishing, f...
5	1rth86s	[D] Reported our meta-reviewer in this ARR cyc...	[acl, arr, commit, cycle, doc, explaining, mar...
6	1rt8t19	[D] Has interpretability research been applied...	[applied, interpretability, model, research, t...
7	1rt4lyd	[D] ran controlled experiments on meta's COCON...	[actually, coconut, controlled, experiments, f...
8	1rsxqoi	[R] biomarker peak detection using machine lea...	[biomarker, collaborate, detection, learning, ...
9	1rsxlg8	[P] ColQwen3.5-v2 4.5B is out!	[out]

In [7]:

```
# Create a bipartite edge list:
# one edge means one post contains one keyword

edge_rows = []

for _, row in posts_df.iterrows():
    post_id = row["post_id"]
    for kw in row["keywords"]:
        edge_rows.append({
```

```
            "post_id": post_id,
            "keyword": kw
        })

post_keyword_edges_df = pd.DataFrame(edge_rows)

print("Total post-keyword edges:", len(post_keyword_edges_df))
post_keyword_edges_df.head(10)
```

```
Total post-keyword edges: 792
```

Out[7]:

	post_id	keyword
0	1rtsbkv	autoresearch
1	1rtsbkv	database
2	1rtsbkv	evolutionary
3	1rtsbkv	karpathy
4	1rtouwo	anyone
5	1rtouwo	deep
6	1rtouwo	dive
7	1rtouwo	interested
8	1rtouwo	neuroscience
9	1rtouwo	series

In [8]:

```
post_keyword_edges_df.to_csv("reddit_post_keyword_edges.csv", index=False)
print("Saved: reddit_post_keyword_edges.csv")
```

```
Saved: reddit_post_keyword_edges.csv
```

RQ1: What are the most frequently discussed topics in Reddit Machine Learning posts?

In [9]:

```
# Show most frequent keywords
all_keywords = [kw for keywords in posts_df["keywords"] for kw in keywords]
keyword_counts = Counter(all_keywords)

top_keywords_df = pd.DataFrame(
    keyword_counts.items(),
    columns=["keyword", "frequency"]
).sort_values(by="frequency", ascending=False)

print("Top 20 most frequent keywords:")
top_keywords_df.head(20)
```

```
Top 20 most frequent keywords:
```

Out[9]:

	keyword	frequency
95	open	9
94	llm	8
100	source	6
169	based	6
65	actually	5
179	models	5
62	model	5
15	arr	5
128	papers	5
13	acl	4
77	reasoning	4
64	training	4
27	arxiv	4
116	paper	4
63	research	4
440	device	4
4	anyone	4
61	interpretability	3
56	reviews	3
83	learning	3

In [10]:

```
# If two keywords appear in the same post title,
# they are connected in the keyword network.

keyword_pair_counts = Counter()

for keywords in posts_df["keywords"]:
    unique_keywords = sorted(set(keywords))
    if len(unique_keywords) >= 2:
        for pair in combinations(unique_keywords, 2):
            keyword_pair_counts[pair] += 1

keyword_network_rows = []

for (kw1, kw2), weight in keyword_pair_counts.items():
    keyword_network_rows.append({
        "source": kw1,
        "target": kw2,
```

```
        "weight": weight
    })

keyword_network_edges_df = pd.DataFrame(keyword_network_rows)

print("Total keyword-keyword edges:", len(keyword_network_edges_df))
keyword_network_edges_df.head(10)
```

```
Total keyword-keyword edges: 3386
```

Out[10]:

	source	target	weight
0	autoresearch	database	1
1	autoresearch	evolutionary	1
2	autoresearch	karpathy	1
3	database	evolutionary	1
4	database	karpathy	1
5	evolutionary	karpathy	1
6	anyone	deep	1
7	anyone	dive	1
8	anyone	interested	1
9	anyone	neuroscience	1

In [11]:

```
# Keep only stronger keyword connections
# We can change the threshold if needed

filtered_keyword_edges_df = keyword_network_edges_df[
    keyword_network_edges_df["weight"] >= 2
].reset_index(drop=True)

print("Filtered keyword-keyword edges:", len(filtered_keyword_edges_df))
filtered_keyword_edges_df.head(10)
```

```
Filtered keyword-keyword edges: 39
```

Out[11]:

	source	target	weight
0	acl	arr	4
1	acl	cycle	2
2	acl	jan	2
3	arr	cycle	2
4	arr	jan	2
5	arr	meta	2
6	arr	reviews	2
7	meta	reviews	2
8	interpretability	model	2
9	model	training	2

In [12]:

```
filtered_keyword_edges_df.to_csv("reddit_keyword_network_edges.csv", index=Fal
print("Saved: reddit_keyword_network_edges.csv")
```

```
Saved: reddit_keyword_network_edges.csv
```

In [13]:

```
# Build keyword graph
G = nx.from_pandas_edgelist(
    filtered_keyword_edges_df,
    source="source",
    target="target",
    edge_attr="weight",
    create_using=nx.Graph()
)

print("Number of keyword nodes:", G.number_of_nodes())
print("Number of keyword edges:", G.number_of_edges())
```

```
Number of keyword nodes: 35
Number of keyword edges: 39
```

In [14]:

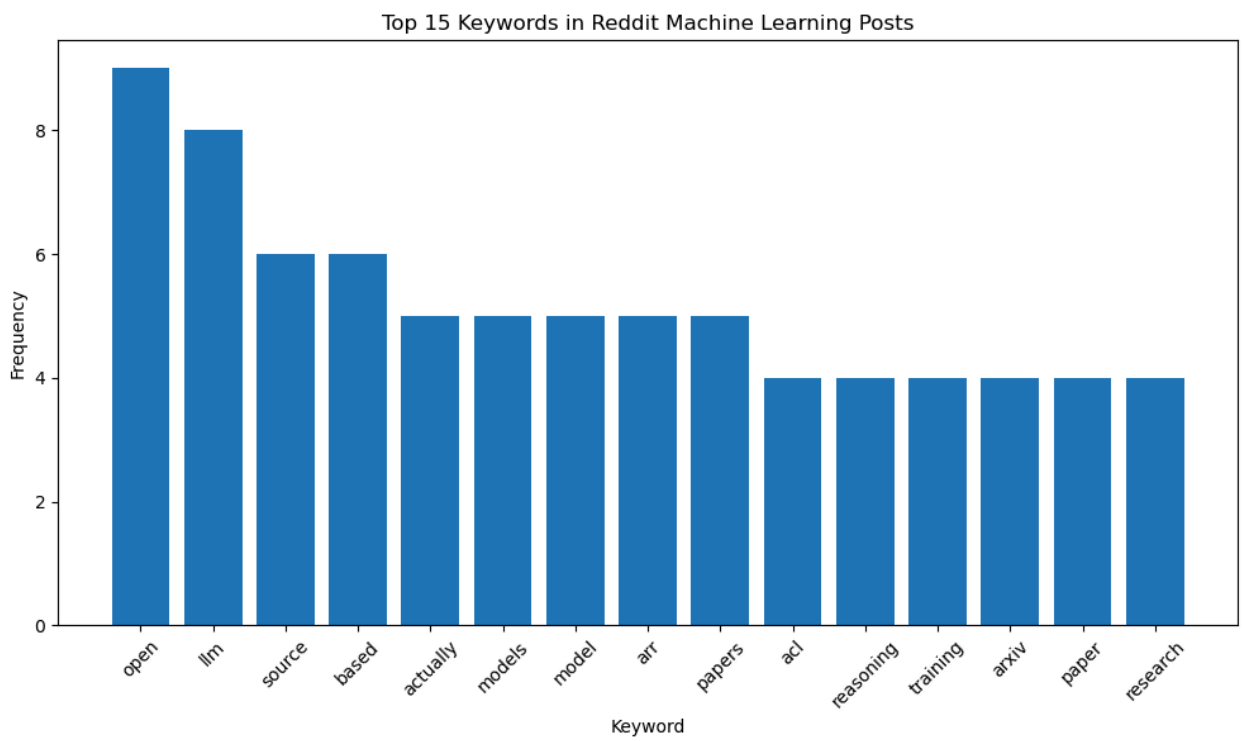
```
# Basic network statistics
num_nodes = G.number_of_nodes()
num_edges = G.number_of_edges()
density = nx.density(G)

# Average degree
degrees = dict(G.degree())
avg_degree = sum(degrees.values()) / num_nodes if num_nodes > 0 else 0

print("Network Statistics")
print("------------------")
print("Nodes:", num_nodes)
print("Edges:", num_edges)
print("Density:", density)
```

```
print("Average Degree:", avg_degree)
```

```
Network Statistics
------------------
Nodes: 35
Edges: 39
Density: 0.06554621848739496
Average Degree: 2.2285714285714286
```

In [15]:

```
# Visualize keyword frequency distribution
plt.figure(figsize=(10, 6))
plt.bar(top_keywords_df["keyword"].head(15), top_keywords_df["frequency"].head
plt.title("Top 15 Keywords in Reddit Machine Learning Posts")
plt.xlabel("Keyword")
plt.ylabel("Frequency")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



RQ2: Which keywords are the most influential in the Machine Learning discussion network?

In [16]:

```
# Community detection
communities = list(greedy_modularity_communities(G))

print("Number of communities detected:", len(communities))
print()

for i, community in enumerate(communities, start=1):
    sample_nodes = list(community)[:10]
    print(f"Community {i}: size = {len(community)}")
```

```
    print("Sample keywords:", sample_nodes)
    print()
```

```
Number of communities detected: 9

Community 1: size = 8
Sample keywords: ['source', 'project', 'tool', 'open', 'llms', 'leaderboard',
'analysis', 'reasoning']

Community 2: size = 7
Sample keywords: ['real', 'time', 'benchmarking', 'scoring', 'llm', 'papers',
'actually']

Community 3: size = 6
Sample keywords: ['reviews', 'acl', 'jan', 'cycle', 'arr', 'meta']

Community 4: size = 3
Sample keywords: ['training', 'interpretability', 'model']

Community 5: size = 3
Sample keywords: ['tts', 'device', 'swift']

Community 6: size = 2
Sample keywords: ['code', 'paper']

Community 7: size = 2
Sample keywords: ['cvpr', 'workshop']

Community 8: size = 2
Sample keywords: ['fine', 'tuning']

Community 9: size = 2
Sample keywords: ['combining', 'execution']
```

In [17]:

```
# Create community mapping
community_map = {}

for i, community in enumerate(communities, start=1):
    for node in community:
        community_map[node] = i

community_df = pd.DataFrame({
    "keyword": list(community_map.keys()),
    "community": list(community_map.values())
})

community_df.head(20)
```

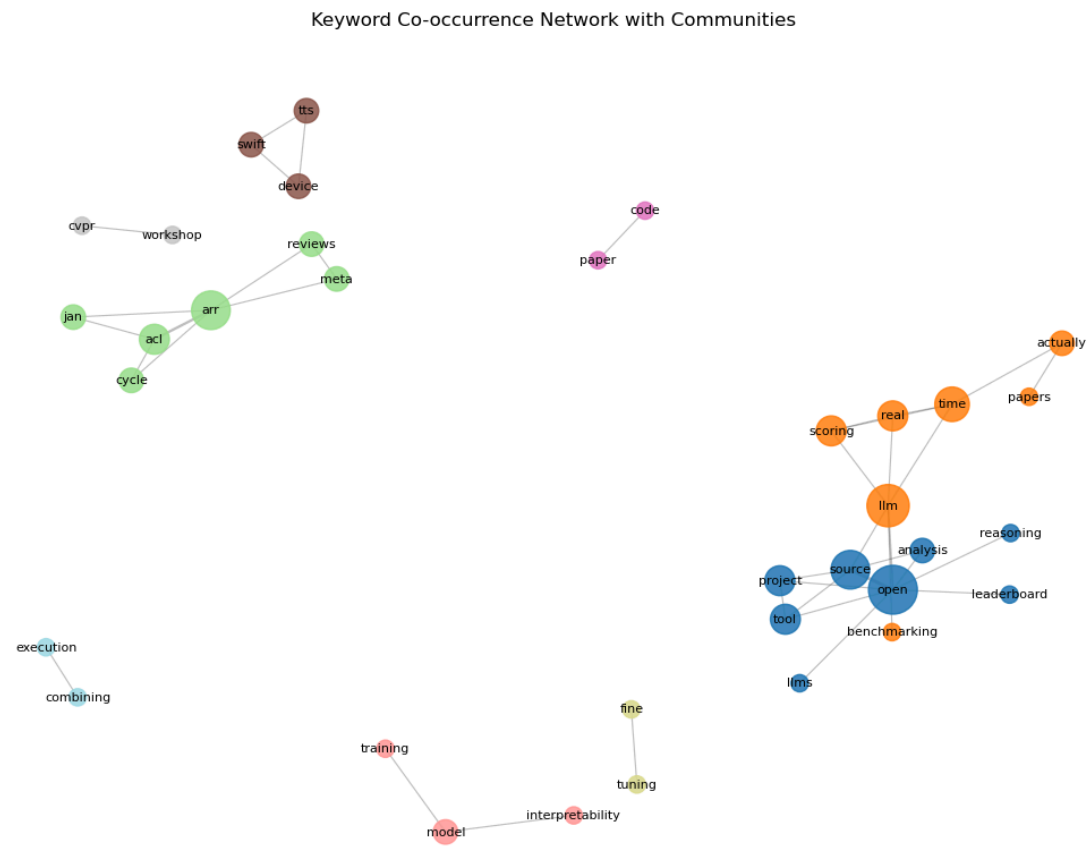
Out[17]:

	keyword	community
0	source	1
1	project	1
2	tool	1
3	open	1
4	llms	1
5	leaderboard	1
6	analysis	1
7	reasoning	1
8	real	2
9	time	2
10	benchmarking	2
11	scoring	2
12	llm	2
13	papers	2
14	actually	2
15	reviews	3
16	acl	3
17	jan	3
18	cycle	3
19	arr	3

In [18]:

```
# Visualize keyword network with communities
plt.figure(figsize=(14, 10))

pos = nx.spring_layout(G, seed=42, k=0.5)

node_colors = [community_map[node] for node in G.nodes()]
node_sizes = [G.degree(node) * 120 for node in G.nodes()]

nx.draw_networkx_nodes(
    G,
    pos,
    node_size=node_sizes,
    node_color=node_colors,
    cmap=plt.cm.tab20,
    alpha=0.85
)
```

```
edge_widths = [G[u][v]["weight"] * 0.4 for u, v in G.edges()]
nx.draw_networkx_edges(
    G,
    pos,
    width=edge_widths,
    alpha=0.25
)

nx.draw_networkx_labels(
    G,
    pos,
    font_size=8
)

plt.title("Keyword Co-occurrence Network with Communities")
plt.axis("off")
plt.show()
```



In [19]:
```
# Degree centrality
degree_centrality = nx.degree_centrality(G)

degree_centrality_df = pd.DataFrame(
    degree_centrality.items(),
    columns=["keyword", "degree_centrality"]
).sort_values(by="degree_centrality", ascending=False)
```

```
print("Top 15 keywords by degree centrality:")
degree_centrality_df.head(15)
```

```
Top 15 keywords by degree centrality:
```

Out[19]:

	keyword	degree_centrality
11	open	0.235294
10	llm	0.176471
14	source	0.147059
1	arr	0.147059
15	time	0.117647
0	acl	0.088235
18	tool	0.088235
16	project	0.088235
12	real	0.088235
13	scoring	0.088235
33	swift	0.058824
32	device	0.058824
27	analysis	0.058824
26	actually	0.058824
34	tts	0.058824

In [20]:

```
# Betweenness centrality
betweenness_centrality = nx.betweenness_centrality(G)

betweenness_df = pd.DataFrame(
    betweenness_centrality.items(),
    columns=["keyword", "betweenness_centrality"]
).sort_values(by="betweenness_centrality", ascending=False)

print("Top 15 keywords by betweenness centrality:")
betweenness_df.head(15)
```

```
Top 15 keywords by betweenness centrality:
```

Out[20]:

	keyword	betweenness_centrality
10	llm	0.094474
11	open	0.084670
15	time	0.042781
26	actually	0.023173
14	source	0.020499
1	arr	0.011586
7	model	0.001783
0	acl	0.000891
28	combining	0.000000
27	analysis	0.000000
21	cvpr	0.000000
29	execution	0.000000
30	llms	0.000000
25	leaderboard	0.000000
31	papers	0.000000

In [21]:

```
# PageRank
pagerank_scores = nx.pagerank(G)

pagerank_df = pd.DataFrame(
    pagerank_scores.items(),
    columns=["keyword", "pagerank"]
).sort_values(by="pagerank", ascending=False)

print("Top 15 keywords by PageRank:")
pagerank_df.head(15)
```

```
Top 15 keywords by PageRank:
```

Out[21]:

	keyword	pagerank
11	open	0.081571
1	arr	0.054135
10	llm	0.053979
14	source	0.050317
7	model	0.041691
15	time	0.037055
0	acl	0.036352
20	paper	0.028571
21	cvpr	0.028571
23	fine	0.028571
22	workshop	0.028571
24	tuning	0.028571
28	combining	0.028571
29	execution	0.028571
32	device	0.028571

In [22]:

```
# Combine all centrality measures
centrality_summary_df = degree_centrality_df.merge(
    betweenness_df,
    on="keyword"
).merge(
    pagerank_df,
    on="keyword"
)

centrality_summary_df = centrality_summary_df.sort_values(
    by="pagerank",
    ascending=False
).reset_index(drop=True)

centrality_summary_df.head(20)
```

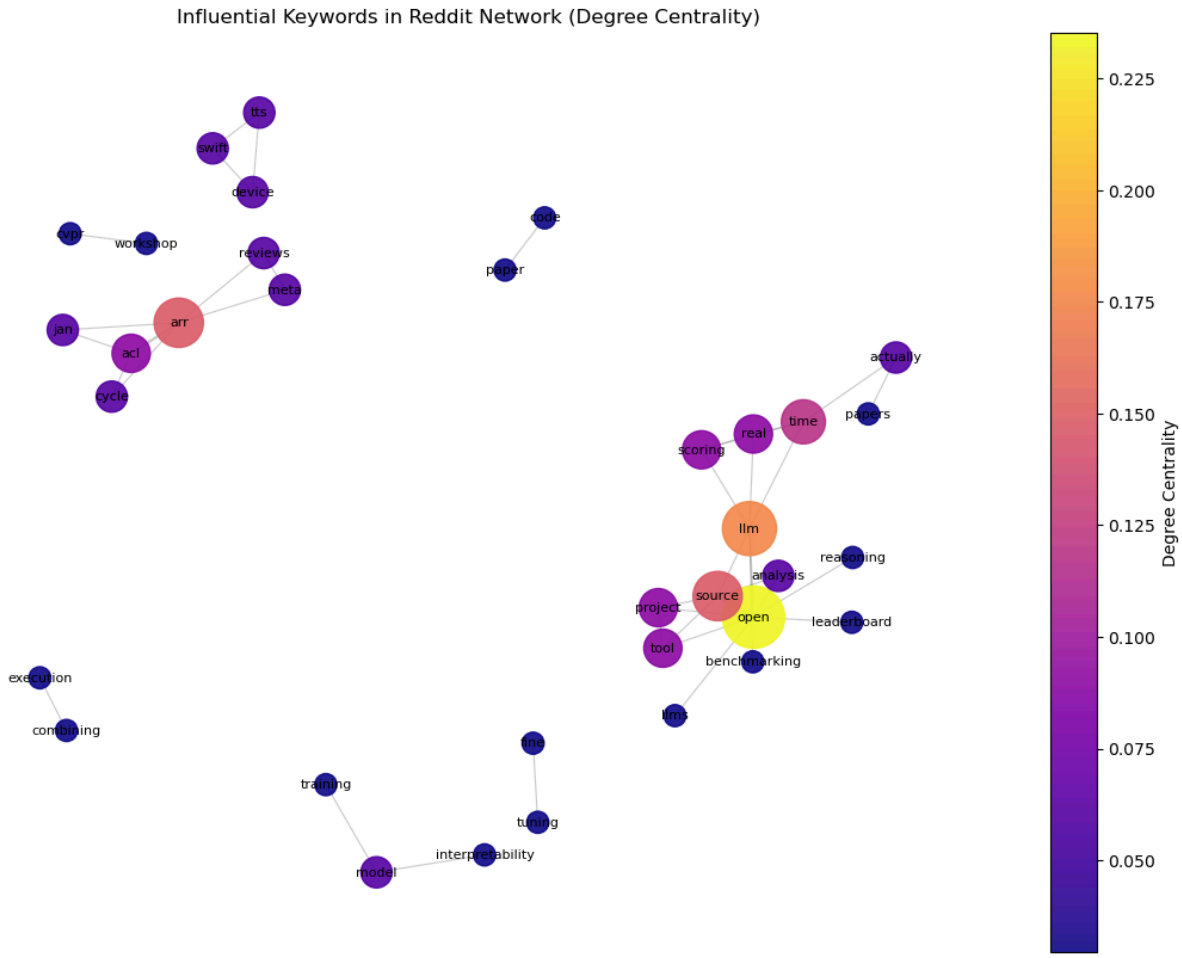
Out[22]:

	keyword	degree_centrality	betweenness_centrality	pagerank
0	open	0.235294	0.084670	0.081571
1	arr	0.147059	0.011586	0.054135
2	llm	0.176471	0.094474	0.053979
3	source	0.147059	0.020499	0.050317
4	model	0.058824	0.001783	0.041691
5	time	0.117647	0.042781	0.037055
6	acl	0.088235	0.000891	0.036352
7	tuning	0.029412	0.000000	0.028571
8	paper	0.029412	0.000000	0.028571
9	execution	0.029412	0.000000	0.028571
10	code	0.029412	0.000000	0.028571
11	combining	0.029412	0.000000	0.028571
12	tts	0.058824	0.000000	0.028571
13	device	0.058824	0.000000	0.028571
14	fine	0.029412	0.000000	0.028571
15	workshop	0.029412	0.000000	0.028571
16	swift	0.058824	0.000000	0.028571
17	cvpr	0.029412	0.000000	0.028571
18	scoring	0.088235	0.000000	0.026817
19	real	0.088235	0.000000	0.026817

In [23]:

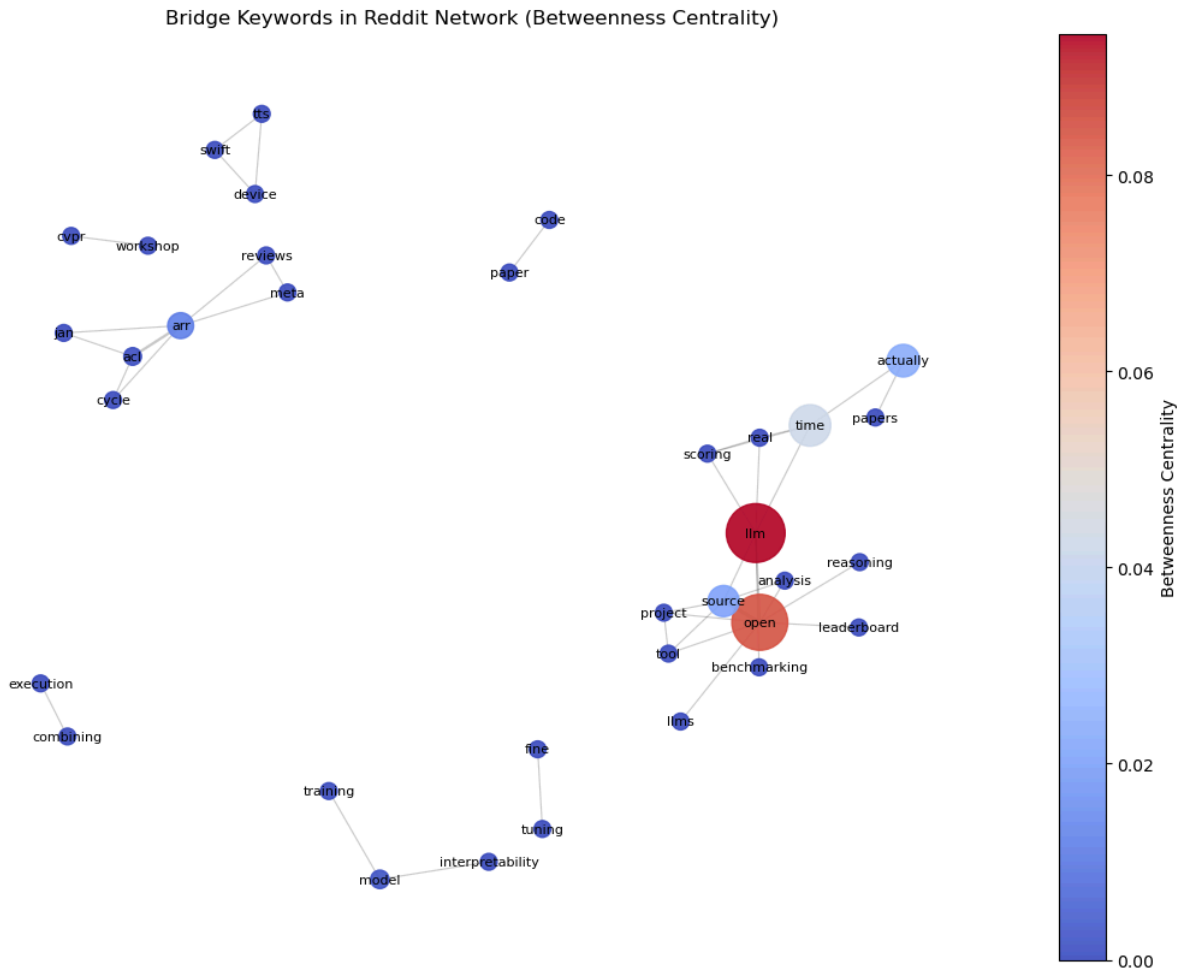
```
# Visualize degree centrality network
plt.figure(figsize=(14, 10))

pos = nx.spring_layout(G, seed=42, k=0.5)

node_sizes = [degree_centrality[node] * 6000 for node in G.nodes()]
node_colors = [degree_centrality[node] for node in G.nodes()]

nodes = nx.draw_networkx_nodes(
    G,
    pos,
    node_size=node_sizes,
    node_color=node_colors,
    cmap=plt.cm.plasma,
    alpha=0.9
)
```

```
edge_widths = [G[u][v]["weight"] * 0.4 for u, v in G.edges()]
nx.draw_networkx_edges(G, pos, width=edge_widths, alpha=0.2)
nx.draw_networkx_labels(G, pos, font_size=8)

plt.colorbar(nodes, label="Degree Centrality")
plt.title("Influential Keywords in Reddit Network (Degree Centrality)")
plt.axis("off")
plt.show()
```



In [24]:
```
# Visualize betweenness centrality network
plt.figure(figsize=(14, 10))

pos = nx.spring_layout(G, seed=42, k=0.5)

node_sizes = [betweenness_centrality[node] * 12000 + 100 for node in G.nodes()
node_colors = [betweenness_centrality[node] for node in G.nodes()]

nodes = nx.draw_networkx_nodes(
    G,
    pos,
    node_size=node_sizes,
    node_color=node_colors,
    cmap=plt.cm.coolwarm,
    alpha=0.9
```

```
)

edge_widths = [G[u][v]["weight"] * 0.4 for u, v in G.edges()]
nx.draw_networkx_edges(G, pos, width=edge_widths, alpha=0.2)
nx.draw_networkx_labels(G, pos, font_size=8)

plt.colorbar(nodes, label="Betweenness Centrality")
plt.title("Bridge Keywords in Reddit Network (Betweenness Centrality)")
plt.axis("off")
plt.show()
```

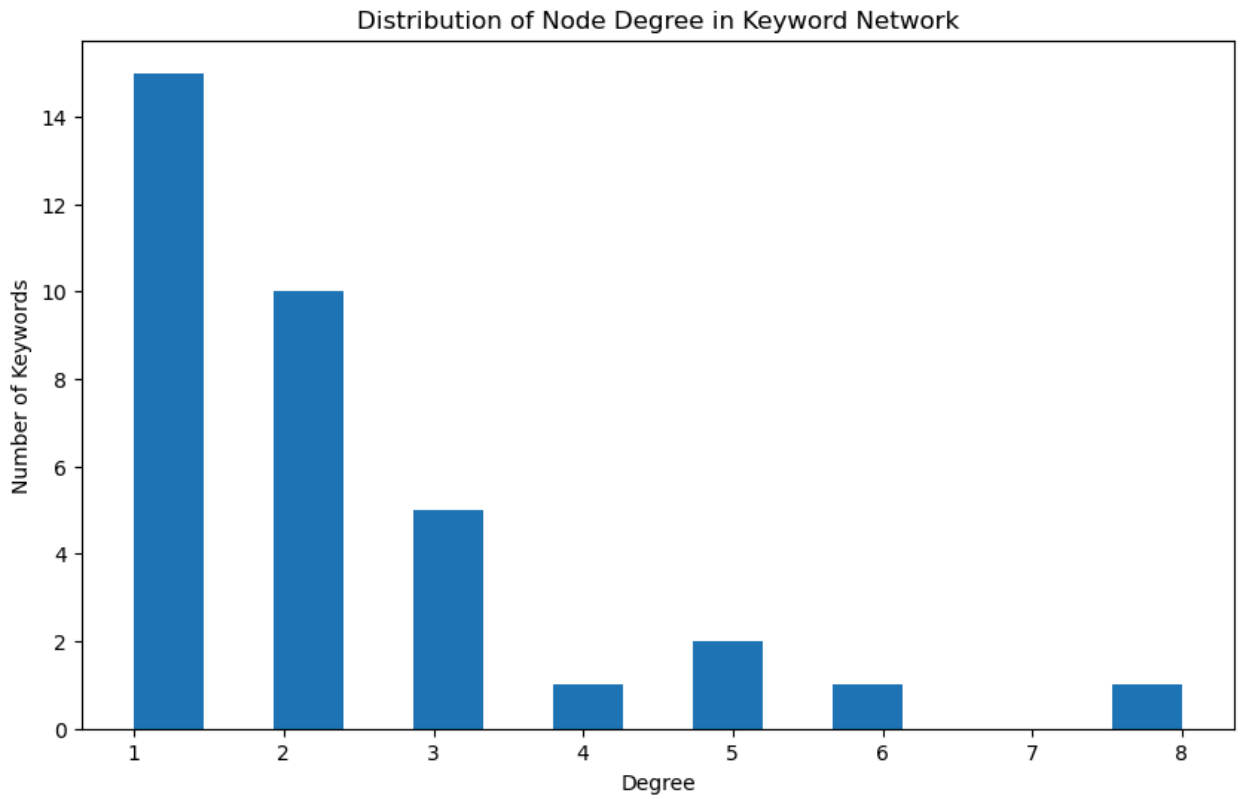


In [25]:
```
# Visualize PageRank network
plt.figure(figsize=(14, 10))

pos = nx.spring_layout(G, seed=42, k=0.5)

node_sizes = [pagerank_scores[node] * 20000 for node in G.nodes()]
node_colors = [pagerank_scores[node] for node in G.nodes()]

nodes = nx.draw_networkx_nodes(
    G,
    pos,
    node_size=node_sizes,
    node_color=node_colors,
```

```
    cmap=plt.cm.viridis,
    alpha=0.9
)

edge_widths = [G[u][v]["weight"] * 0.4 for u, v in G.edges()]
nx.draw_networkx_edges(G, pos, width=edge_widths, alpha=0.2)
nx.draw_networkx_labels(G, pos, font_size=8)

plt.colorbar(nodes, label="PageRank Score")
plt.title("Influential Keywords Based on PageRank")
plt.axis("off")
plt.show()
```

In [26]:

```
# Degree distribution histogram
plt.figure(figsize=(10, 6))
plt.hist(list(degrees.values()), bins=15)
plt.title("Distribution of Node Degree in Keyword Network")
plt.xlabel("Degree")
plt.ylabel("Number of Keywords")
plt.show()
```



In [27]:
```
# Average clustering coefficient
avg_clustering = nx.average_clustering(G)
print("Average clustering coefficient:", avg_clustering)
```

```
Average clustering coefficient: 0.41176870748299316
```

In [28]:
```
# Top weighted keyword pairs
top_weighted_edges_df = filtered_keyword_edges_df.sort_values(
    by="weight",
    ascending=False
).reset_index(drop=True)

print("Top 20 strongest keyword pairs:")
top_weighted_edges_df.head(20)
```

```
Top 20 strongest keyword pairs:
```

Out[28]:

	source	target	weight
0	open	source	5
1	acl	arr	4
2	llm	open	3
3	project	tool	2
4	real	time	2
5	scoring	time	2
6	source	tool	2
7	code	paper	2
8	cvpr	workshop	2
9	fine	tuning	2
10	leaderboard	open	2
11	actually	time	2
12	analysis	open	2
13	analysis	source	2
14	combining	execution	2
15	llms	open	2
16	actually	papers	2
17	device	swift	2
18	device	tts	2
19	real	scoring	2

In [29]:

```
# Export results tables
top_keywords_df.to_csv("top_keywords_frequency.csv", index=False)
community_df.to_csv("keyword_communities.csv", index=False)
centrality_summary_df.to_csv("keyword_centrality_summary.csv", index=False)

print("Saved result tables:")
print("- top_keywords_frequency.csv")
print("- keyword_communities.csv")
print("- keyword_centrality_summary.csv")
```

```
Saved result tables:
- top_keywords_frequency.csv
- keyword_communities.csv
- keyword_centrality_summary.csv
```

In [30]:

```
# Final interpretation printout
print("Final Summary")
```

```python
print("-------------")
print(f"Posts collected: {len(posts_df)}")
print(f"Keyword nodes: {G.number_of_nodes()}")
print(f"Keyword edges: {G.number_of_edges()}")
print(f"Number of communities: {len(communities)}")
print(f"Average clustering coefficient: {avg_clustering:.4f}")

print("\nTop 10 keywords by PageRank:")
print(centrality_summary_df[["keyword", "pagerank"]].head(10))
```

```
Final Summary
-------------
Posts collected: 100
Keyword nodes: 35
Keyword edges: 39
Number of communities: 9
Average clustering coefficient: 0.4118

Top 10 keywords by PageRank:
     keyword  pagerank
0       open  0.081571
1        arr  0.054135
2        llm  0.053979
3     source  0.050317
4      model  0.041691
5       time  0.037055
6        acl  0.036352
7     tuning  0.028571
8      paper  0.028571
9  execution  0.028571
```

In [ ]: